

Compact expressions for the Biot-Savart fields of a filamentary segment

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Compact and computationally efficient expressions for the Biot-Savart magnetic field and vector potential of a single filamentary segment are derived. The expressions are singular only on the segment.

Computing the magnetic field and magnetic vector potential due to known currents in coils is a much-utilized procedure in plasma physics. The magnetic field is needed as input to magnetohydrodynamic (MHD) equilibrium codes, for tracing magnetic field lines, and for following particle orbits. The magnetic vector potential is used for computing magnetic fluxes and inductances.

One common technique¹⁻⁴ is to approximate a coil as a sequence of straight line segments, connected end-to-end to form a closed polygon in space. The magnetic field and vector potential due to a filamentary segment can be computed analytically, and the contributions from each segment are added together numerically to obtain the net field and potential due to an arbitrarily complex coil. Coils with substantial cross-sectional areas can be approximated with multiple non-intersecting polygons.

Using filamentary segments is a very flexible technique. There are essentially no restrictions on the shape of the coil. The accuracy can be readily improved by increasing the number of segments per polygon, or the number of polygons. The procedure is computationally rapid, and, the resulting magnetic field is well behaved: $\nabla \times \mathbf{B} = 0$ to roundoff accuracy (outside the filaments, when polygons are closed and each segment carries identical current, so that $\nabla \cdot \mathbf{J} = 0$), and $\nabla \cdot \mathbf{B} = 0$, also to roundoff accuracy.

The major disadvantage to using filamentary segments to model coils is that the fields diverge on the filamentary segments, and so computing fields nearby, and in the interior of coils is necessarily inaccurate. When this limitation is important, more accurate results can be obtained with models for the coils made from volume current elements^{3,5-7}.

One difficulty with some of the published expressions for the magnetic field and vector potential of a single filamentary segment is that the expressions are singular not

only on the segment itself, but also on the line extensions containing the segment. This singularity on the segment extension involves a ratio of terms, each of which evaluates to zero, but for which L'Hôpital's rule yields a finite ratio. If this numerical difficulty is recognized, a test can be made to check if the evaluation point is sufficiently near the segment extension, and an alternate expression can be used to evaluate the ratio^{2,8}. If the difficulty is not recognized, there is significant loss of numerical accuracy near the segment extension. The problem exists in the magnetic field expressions in Refs. (1,3,4) and in the vector potential expression in Refs. (2-4). (Reference 1 does not give an expression for the vector potential.) In Ref. 8 an alternate expression for the magnetic field that is singular on the segment perpendicular bisecting plane is given. The difficulty does not arise with the magnetic field expression of Ref. 2. Their result is singular only on the segment itself (as it should be), and is well behaved everywhere else.

In the next section we derive expressions for both the magnetic field and the vector potential that are singular only on the segment itself. The derived expressions are good for rapid numerical evaluation of the fields. The magnetic field expression is computationally more efficient than that of Ref. 2.

Figure 1 shows a single filamentary segment that starts at position $\mathbf{x}_i$ and extends to $\mathbf{x}_f$. The location where the fields are to be calculated is denoted $\mathbf{x}$, and we define

$L \equiv |\mathbf{x}_f - \mathbf{x}_i|$ (length of the segment), $\hat{\mathbf{e}} \equiv (\mathbf{x}_f - \mathbf{x}_i)/L$ (unit vector along the segment) and

$\mathbf{R}_{i(f)} \equiv \mathbf{x} - \mathbf{x}_{i(f)}$ (vector from segment end points to the observation point $\mathbf{x}$),

$R_{i(f)} \equiv |\mathbf{x} - \mathbf{x}_{i(f)}|$, $R_{i(f)\parallel} \equiv \hat{\mathbf{e}} \cdot \mathbf{R}_{i(f)}$. The Biot-Savart expression for the vector potential is⁹

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d\mathbf{x}'. \quad (1)$$

Parametrizing the straight-line coil segment as $\mathbf{c}(\varpi) = \mathbf{x}_i + \varpi(\mathbf{x}_f - \mathbf{x}_i)$ for $0 < \varpi < 1$, we

have

$$\mathbf{A}(\mathbf{x}) = \frac{\varpi_0 I}{4\varpi} L \hat{\mathbf{e}} \int_0^1 \frac{d\varpi}{|\mathbf{x} - \mathbf{x}_i - \varpi L \hat{\mathbf{e}}|} = \frac{\varpi_0 I}{4\varpi} L \hat{\mathbf{e}} \int_0^1 \frac{R_i^2 - \varpi^2 L^2 + 2\varpi L R_{i\parallel}}{[R_i^2 - \varpi^2 L^2 + 2\varpi L R_{i\parallel}]^{3/2}} d\varpi. \quad (2)$$

Note that the denominator of the integrand does not vanish except when $\mathbf{x}$ is on the line segment. The integral is standard. Recognizing that

$$L = \hat{\mathbf{e}} \cdot (\mathbf{R}_i - \mathbf{R}_f) = R_{i\parallel} - R_{f\parallel}, \quad (3)$$

we obtain

$$\mathbf{A}(\mathbf{x}) = \frac{\varpi_0 I}{4\varpi} \hat{\mathbf{e}} \ln \left[\frac{R_f - R_{f\parallel}}{R_i - R_{i\parallel}} \right]. \quad (4)$$

This expression evaluates to $\ln[0/0]$ whenever $\mathbf{x}$ is on the forward extension of the segment, that is when $\mathbf{x} = \mathbf{x}_f + \varpi \hat{\mathbf{e}}$, $\varpi > 0$.

To obtain the new expression for the vector potential, note that

$$R_f^2 - R_i^2 = (\mathbf{R}_f - \mathbf{R}_i) \cdot (\mathbf{R}_f + \mathbf{R}_i) = \varpi L (R_{f\parallel} + R_{i\parallel}). \quad (5)$$

Solving Eqs. (3) and (5) for $R_{i\parallel}$ and $R_{f\parallel}$ yields

$$R_{f\parallel} = \frac{R_i^2 - R_f^2}{2L} \pm \frac{L}{2}. \quad (6)$$

Substituting these into Eq. (4) yields the desired expression:

$$\mathbf{A}(\mathbf{x}) = \frac{\varpi_0 I}{4\varpi} \hat{\mathbf{e}} \ln \left[\frac{R_i + R_f + L}{R_i + R_f - L} \right] = \frac{\varpi_0 I}{4\varpi} \hat{\mathbf{e}} f(\varpi). \quad (7)$$

where $f(\varpi) \equiv \ln[(1 + \varpi)/(1 - \varpi)]$ and $\varpi \equiv L/(R_i + R_f)$. The common factor $(L + R_f - R_i)$ that vanishes on the forward line segment extension has cancelled top and bottom. It is

interesting to note that e is the eccentricity of the ellipse through the point $\mathbf{x}$ with foci at $\mathbf{x}_i$ and $\mathbf{x}_f$.

The magnetic field can be obtained by a similar integration over the straight-line curve. Here we derive the magnetic field by using $\mathbf{B} = \nabla \times \mathbf{A}$:

$$\begin{aligned} \frac{4\pi}{\mu_0 I} \mathbf{B} &= \nabla f(\mathbf{r}) \times \hat{\mathbf{e}} = f'(\mathbf{r}) \frac{L}{(R_i + R_f)^2} \hat{\mathbf{e}} \times (\mathbf{r} - \mathbf{r}_i) \\ &= f'(\mathbf{r}) \frac{L}{L} \hat{\mathbf{e}} \times \left(\frac{\mathbf{r}_i}{R_i} + \frac{\mathbf{r}_f}{R_f} \right) = \hat{\mathbf{e}} \times \mathbf{r}_i \frac{f'(\mathbf{r})}{R_i R_f} \end{aligned}$$

where we have used $\nabla \cdot \hat{\mathbf{e}} = 0$ and $\mathbf{r}_f = \mathbf{r}_i + L\hat{\mathbf{e}}$. Noting that $f'(\mathbf{r}) = 2/(1 - e^2)$ we obtain:

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \hat{\mathbf{e}} \times \mathbf{r}_i \frac{2\pi}{1 - e^2} \frac{1}{R_i R_f} = \frac{\mu_0 I}{4\pi} \hat{\mathbf{e}} \times \mathbf{r}_i \frac{2L(R_i + R_f)}{R_i R_f} \frac{1}{(R_i + R_f)^2 \pi L^2} \quad (8)$$

Equations (7) and (8) are the main results of this paper. These expressions diverge only when $\mathbf{x}$ is on the filamentary segment, where $L = R_i + R_f$ and $e=1$. They offer significant improvements in computational efficiency compared with many previously published expressions: no trigonometric functions are required, and avoiding singularities involves a simple test on e .

Figure 2 shows the numerical error resulting from computing the vector potential using Eq. (4), near the forward extension of a filamentary segment. Equation 4 is evaluated at points $\mathbf{x} = (\sqrt{y}, y, 0)$. The relative error increases as y decreases below 0.1. When y is less than about 10^{-7} (square root of the machine epsilon) the numerical evaluation of Eq. (4) fails with a floating point exception, due to the $\ln(0/0)$ singularity. Evaluation of Eq. (7) has no such difficulty, and is accurate to near machine precision.

In our notation, the expression of Ref. 2 for the magnetic field is

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \mathbf{R}_i \times \mathbf{R}_f \frac{(R_i + R_f)}{R_i R_f (R_i R_f + \mathbf{R}_i \cdot \mathbf{R}_f)}. \quad (9)$$

Equations (8) and (9) are seen to be equivalent when one recognizes that

$(\mathbf{R}_i \times \mathbf{R}_f)^2 = L^2 = R_i^2 \times 2\mathbf{R}_i \cdot \mathbf{R}_f + R_f^2$. Equation (8) should be somewhat more efficient to evaluate than Eq. (9), as the dot product $\mathbf{R}_i \cdot \mathbf{R}_f$ does not need to be computed¹⁰.

The power series expansion of $f(L)$ is

$$f(L) = 2 \sum_{i=0}^{\infty} \frac{L^{2i+1}}{2i+1} = 2 \left[\frac{L}{1} + \frac{L^3}{3} + \frac{L^5}{5} + \frac{L^7}{7} + \dots \right] \quad (9)$$

Since only odd powers of L are present, the series converges rapidly for small L . In applications where rapid computation of the vector potential is important, and most evaluation points are relatively far from the coils (so that most L values are small), evaluating $f(L)$ using the power series for small L and only calling the logarithm for large L may be an effective strategy. If high accuracy is not needed near the coils, then a Padé approximation to $f(L)$, avoiding the logarithm altogether, may be appropriate.

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FIGURES

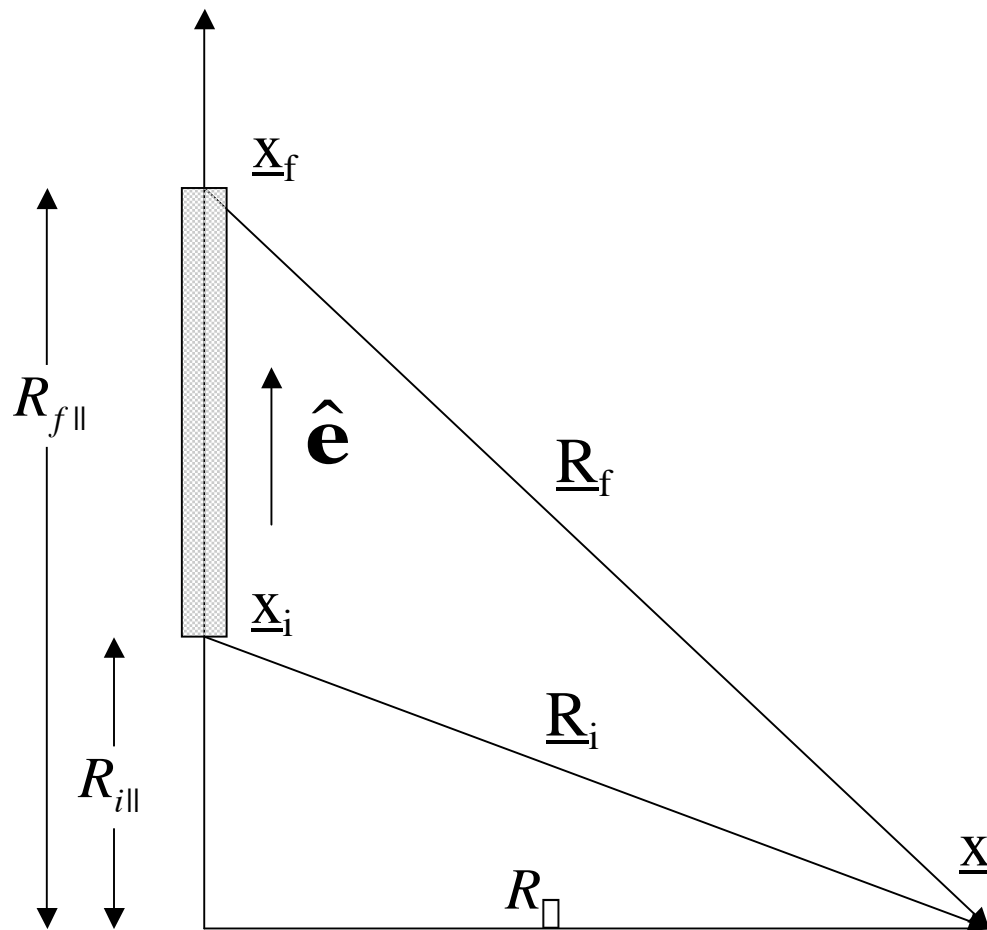


FIGURE 1. A single straight filamentary segment, extending from $\mathbf{x}_i$ to $\mathbf{x}_f$.

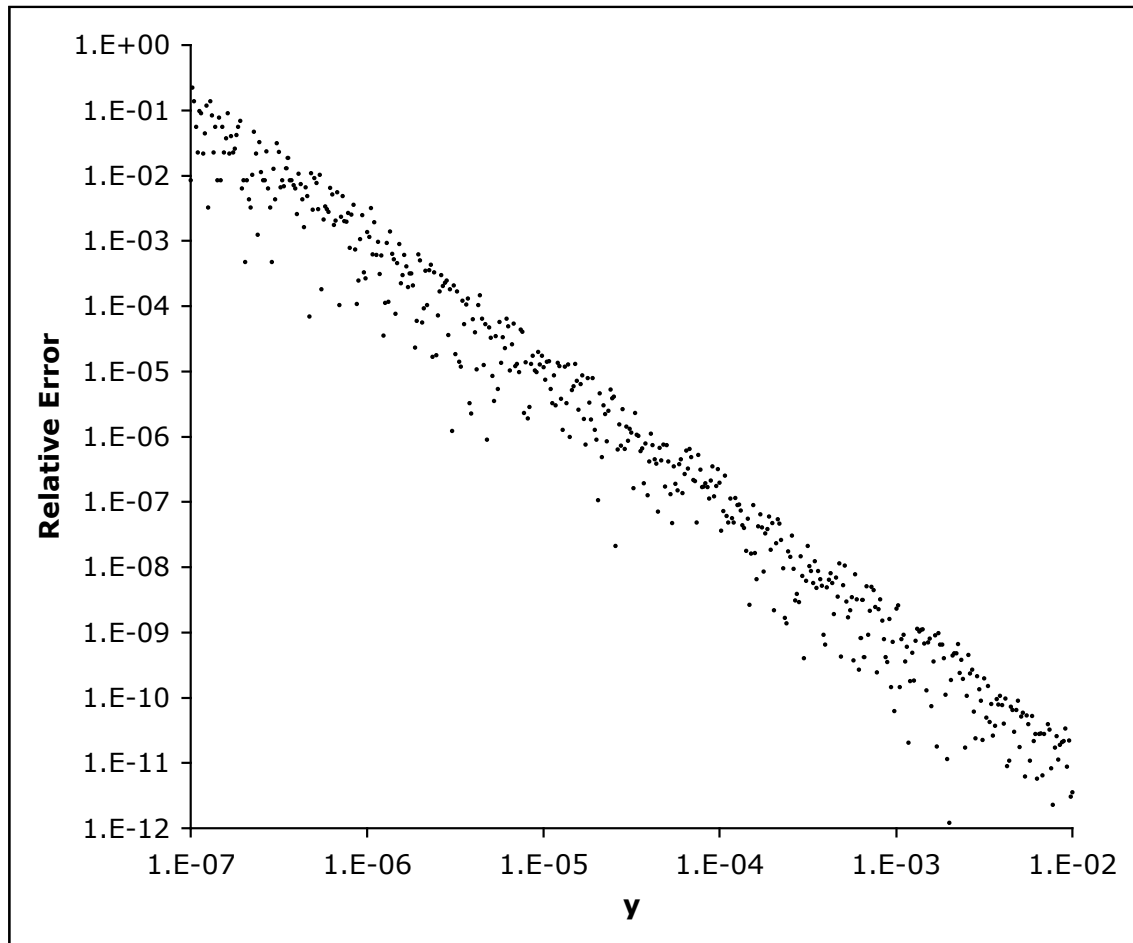


FIGURE 2. Relative error in the magnitude of the vector potential expression of Eq. (4).

A single filamentary segment extends along the x axis from 0 to 1. The vector potential is computed near the forward extension of the segment, at points $\mathbf{x} = (\sqrt{5}, y, 0)$.